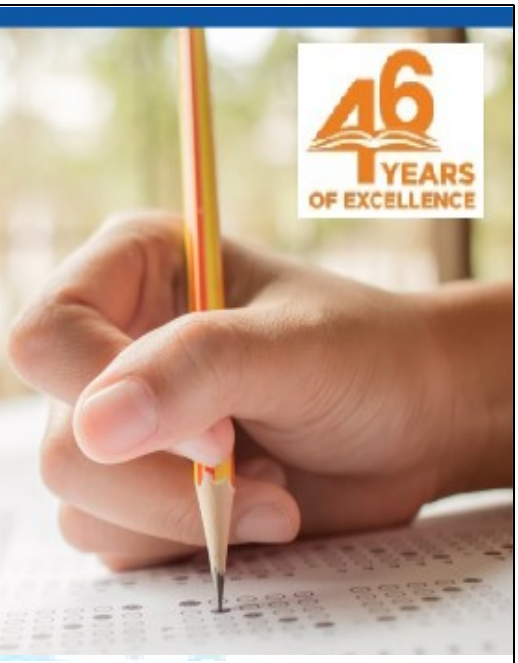
NARAYANA IIT ACADEMY INDIA

JEE ADVANCED GRAND TEST SERIES



46 YEARS OF EXCELLENCE

NARAYANA IIT - JEE ACADEMY - INDIA

Sec: OSR.IIT _*CO-SC(MODEL-A&APEX) Date: 26-04-26
Time: 3 Hrs Max. Marks: 180

26-04-26_OSR.IIT_*CO-SC(MODEL-A&APEX)_JEE ADV_GTA-9(P2)_SYLLABUS

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MATHEMATICS: TOTAL SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

THE NARAYANA GROUP

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TIME:3hrs **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 6)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II (Q.N : 7 – 12)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 + 2 = 6Q)	+2	0	6	12
Sec – III (Q.N : 13 – 16)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	+3	–1	4	12
Sec – IV (Q.N : 17 – 19)	Questions with Non-Negative Integer type	+4	0	3	12
Total				**19**	**60**

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 20 – 25)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II (Q.N : 26 – 31)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 + 2 = 6Q)	+2	0	6	12
Sec – III (Q.N : 32 – 35)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	+3	–1	4	12
Sec – IV (Q.N : 36 – 38)	Questions with Non-Negative Integer type	+4	0	3	12
Total				**19**	**60**

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 39 – 44)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II (Q.N : 45 – 50)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 + 2 = 6Q)	+2	0	6	12
Sec – III (Q.N : 51 – 54)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	+3	–1	4	12
Sec – IV (Q.N : 55 – 57)	Questions with Non-Negative Integer type	+4	0	3	12
Total				**19**	**60**

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MATHEMATICS **MAX.MARKS: 60**

SECTION - 1 (Maximum Marks : 24)

This section contains **SIX (06)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

1. There are 3 apples of which 1 is rotten, 4 oranges of which 2 are rotten and 2 pears of which 1 is rotten. Number of ways to arrange them in a straight line such that (rotten oranges are considered similar and non-rotten fruits of same variety are considered similar):

A) any two rotten fruits have atleast 2 fruits in between them is 0
B) all apples, oranges and pears are together respectively is 216
C) any 2 rotten fruits have exactly 1 fruit between them is 1080
D) all rotten fruits are together is 2160

2. $\sum_{r=2}^{n}\frac{r^6-3r^4-3r^2-3}{r^3-3r^4+3r^2-1}=f(n)+\frac{g(n)}{(h(n))^3}$ where $f(n),g(n),h(n)$ are linear, cubic and quadratic in ' n', respectively, then

A) $f(1)=-\frac{5}{4}$
B) $g(1)=9$
C) $h(1)=2$
D) $f(1)+g(1)+h(1)=\frac{39}{4}$

3. Let $\vec{a},\vec{b}$ and $\vec{c}$ be three unit vectors such that $|\vec{a}+\vec{b}+\vec{c}|=\sqrt{3}$ and let $(\vec{a}\times\vec{b})\cdot(\vec{b}\times\vec{c})_{+}(\vec{b}\times\vec{c})\cdot(\vec{c}\times\vec{a})_{+}(\vec{c}\times\vec{a})\cdot(\vec{a}\times\vec{b})=\lambda$. Which of the following is/are CORRECT?

A) The maximum value of λ is 0
B) If λ is maximum, then volume of parallelepiped determined by $\vec{a},\vec{b}$ and $\vec{c}$ is $\sqrt{3}$
C) If λ is maximum, then the value of $\left|(2\vec{a}+3\vec{b}+4\vec{c})\cdot(\vec{a}\times\vec{b}+5\vec{b}\times\vec{c}+6\vec{c}\times\vec{a})\right|$ is 32
D) None of these

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4. If P and Q are square matrices of order 2 such that $P+\text{adj}\left(Q^{\top}\right)=\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}$ and $P^{\top}-\text{adj}(Q)=\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$, Then

A) Q is a symmetric matrix

B) $P^n = P \forall n \in N$

C) $\left|P+P^2+P^3+P^4+P^5\right|=0$

D) $\left|Q+Q^2+Q^3+Q^4+Q^5\right|=0$

5. Let Z_1, Z_2 be fixed complex numbers and Z is variable complex number such that $|Z-Z_1|^2+|Z-Z_2|^2=k|Z_1-Z_2|^2\,(k\in R)$, then locus of Z is :

A) A circle with centre at midpoint of $Z_1, Z_2 \forall k>0$

B) A circle with diameter more than $|Z_1-Z_2|$ if $\frac{1}{2}<k<1$

C) A circle with diameter more than $|Z_1-Z_2|$ if $k>1$

D) A point if $k=\frac{1}{2}$

6. Let $S=\left\{a\in R \;\middle|\; a=\sin^{-1}\left(\frac{x^2}{4}+\frac{y^2}{9}\right)+\cos^{-1}\left(\frac{x}{2\sqrt{2}}+\frac{y}{3\sqrt{2}}-2\right); x,y\in R\right\}$, then

(Consider principal values for inverse trigonometric functions)

A) Number of elements in the set S is 2

B) Number of elements in the set S is more than 2

C) Number of equivalence relations on set S is 1

D) The mean of element/s in set S is $\frac{3\pi}{2}$

SECTION - 2 (Maximum Marks : 12)

This section contains **TWO (3)** Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +2 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 7 and 8:

Let k and n be two natural numbers and let a and b be any two numbers chosen from the set $P=\{1,2,\ldots k, k+1, k+2,\ldots 2k, 2k+1, 2k+2\ldots 3k,\ldots,(n-1)k+2,\ldots nk\}$

7. If $k=5$ and $n=10$ and a and b are distinct numbers, then what is the probability that a^4-b^4 is divisible by 5 .

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8. If $k=3$ and $n=2$, and a and b may be identical or distinct, then what is the probability that there exist real numbers x, y and z such that $x+y+z=a$ and $xy+yz+zx=b$.

Question Stem for Question Nos. 9 and 10:

Suppose $a_1, a_2, a_3, \ldots, a_{100}$ are positive real numbers such that $a_k = \dfrac{ka_{k-1}}{a_{k-1}-(k-1)}$ for $k=2,3,\ldots,100$. Given that $a_{20}=a_{23}$. Then

9. a_{100} is equal to

10. a_{23} is equal to

Question Stem for Question Nos. 11 and 12:

Consider the functions $f(x): R \to R$ and $g(x): R \to R$ such that $f(x)=\dfrac{x^3}{2}+1-x\int_0^x g(t)\,dt$ and $g(x)=x-\int_0^1 f(t)\,dt$.

11. A tangent parallel to $y=g(x)$ is drawn to the curve $y=f(x)$ such that the tangent cuts the coordinate axes at A and B, the area of the triangle OAB is (where O is the origin) R square units, then the value of 144 R is ____

12. Minimum distance between $y=f(x)$ and $x=f(y)$ is of the form $\dfrac{a}{b\sqrt{c}}$ units (a, b & c are natural numbers satisfying $GCD(a,b)=1$ and c is prime). The least value of $a+b+c$ is equal to ____

SECTION - 3 (Maximum Marks : 12)

This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.
Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
For each question, choose the option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.
Negative Marks: −1 In all other cases.

Question Stem for Question Nos. 13 and 14:

Let $I(n,\lambda)=\int_0^{\pi/2} \dfrac{\sin^n x\cos^{2n}x+\lambda\sin^{n+2}x\cos^n x}{\lambda+\sin^n x+\cos^n x}\,dx$, where $n \in$ odd natural numbers and $\lambda \in R^+$.

13. The value of $\dfrac{\sum_{k=0}^{5} I(n,2^k)}{\sum_{k=0}^{7} I(n,3^k)}$ is

A) 0.25 B) 0.75 C) 0.55 D) 0.50

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14. The value of $\sum_{n=1}^{201} n \cdot {}^{n-1}C_{\frac{n-1}{2}} \ell(n,n!)$ is, $n \in$ odd natural numbers

A) 25.25 B) 15.15 C) 50.4 D) 20.25

Question Stem for Question Nos. 15 and 16:

If $f(x) \geq 0 \forall x \in [a,b]$ then area bounded by $y = f(x)$, x -axis between $x = a$ and $x = b$ is given by $\int_a^b f(x)dx$. Answer the following questions.

15. Let $M = \{(x,y) : \sin^3 x \leq y \leq \sin x$ and $0 \leq x \leq \pi\}$

$N = \{(x,y) : \sin^3 x \leq y \leq |\sin x|$ and $\pi \leq x \leq 2\pi\}$ and $S = M \cup N$, then area of the region formed by point in S is :

A) $\frac{15}{4}$ B) 2 C) 4 D) $\frac{17}{4}$

16. Area of region bounded by Y -axis and the curve $\left(\sqrt{x(4-x)} - y\right)\left(\sqrt{x(4-x)} - 4 + y\right) = 0$ is:

A) $8 - 4\pi$ B) $16 - 8\pi$ C) 16 D) None of these

SECTION - 4 (Maximum Marks : 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

17. Let R be a relation on the set $A = \{1,2,3,4,5,6,7\}$ defined as

$R = \{(1,2), (4,6), (5,6), (7,7)\}$. The least number of element in an equivalence reaction R', defined on the set A, which contains R is K, then $K - 8$ is ____

18. Let $p = \lim_{y \to \infty} \left(\frac{2}{y^2} \left(\lim_{z \to \infty} \frac{1}{z^4} \left(\lim_{x \to 0} \frac{\left((y^2+y+1)x^k+1\right)^{z^2+z+1} - \left((z^2+z+1)x^k+1\right)^{y^2+y+1}}{x^{2k}} \right) \right) \right)^y$ where $k \in N$

and $q = \lim_{n \to \infty} \left(\frac{{}^{2n}C_n \cdot n!}{n^n} \right)^{1/n}$ where $n \in N$. The value of $p.q$ is:

19. If P and Q be two invertible matrices of order 2025×2025 such that $|P| = -2025$ and $P^{-1}Q + Q^{-1}P = I$, then $\frac{\sqrt{|Q|}}{5}$ is

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PHYSICS **MAX.MARKS: 60**

SECTION - 1 (Maximum Marks : 24)

This section contains **SIX (06)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
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Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

20. Radon decays to polonium according to the decay scheme given by:

$$^{222}_{88}\text{Rn} \xrightarrow[T_{1/2}=3.8\text{ days}]{\alpha\text{-decay}} {}^{218}_{84}Po\,.$$

The process has a half life of 3.8 days. State which of the following statement(s) will be true in respect of this decay: (Given: $\ln 2 = 0.693$)

A) If the original mass of radon in a closed container was 1 mg , the amount of radon left after 11.4 days will be 0.125 mg .

B) Activity of radon after 7.6 days will be $\frac{A_0}{4}$ where A^0 is original activity of sample at $t = 0$.

C) If the original number of radon in a closed container was 8 atom, the number of radon left after 7.6 day will be 2 atoms.

D) Po is more stable than Rn.

21. A Plano convex lens ($\mu = 1.5$) of focal length 20 cm has its plane side silvered :

A) The radius of curvature of its curved surface is half that of a surface of equiconvex lens of focal length 20 cm made of same material.

B) An object place at 15 cm from it on the axis on the convex side gives rise to an image at a distance of 30 cm from it.

C) An object located at a distance of 20 cm on the axis on the convex side gives rise to an image at 40 cm from it.

D) It acts as a convex mirror

22. In the Young's double slit experiment, the interference pattern is found to have an intensity ratio between bright and dark fringes as 9 . This implies that -

A) the intensities at the screen due to the two slits are 5 units and 4 units respectively

B) the intensities at the screen due to the two slits are 4 units and 1 unit respectively

C) the amplitude ratio is 3

D) the amplitude ratio is 2

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23. In an x-ray tube the voltage applied is 20 kV . The energy required to remove an electron from L shell is 19.9 KeV . In the x-rays emitted by the tube :

A) minimum wavelength will be 62.1 pm

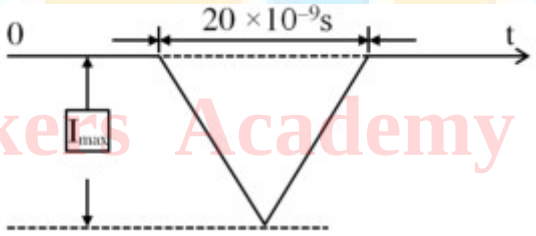
B) energy of the characteristic x-rays will be equal to or less than 19.9 KeV

C) L_α x -ray may be emitted

D) L_α x -ray will have energy 19.9 KeV

24. A photomultiplier tube is to be used to detect light pulses each of which consists of a small but fixed number of photons. The average photoelectric efficiency is 10%. That is photon has 10% probability of causing the emission of a detectable photoelectron. Assume the photoelectrons are multiplied by a factor of 10^6 and that the output current is generated in it. The output current as a function of time can be approximated as shown in figure.

I_{max} when averaged over many pulses is 80μ A . Assume these light pulses to be nonoverlapping in time. Then which of the following are TRUE?



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A) The charge carried in output current by one pulse is 8×10^{-13}C

B) Number of photoelectrons emitted per light pulse is 5

C) Number of photons in one light pulse is 50

D) Number of electrons carried in output current by one pulse is 5×10^5

25. There are two massless springs A and B of spring constant K_A and K_B respectively and $K_A > K_B$, initially in their natural lengths. If W_A and W_B be denoted as work done on A and work done on B respectively, then -

A) If they are compressed by same compression, $W_A > W_B$

B) If they are compressed by same force (upto equilibrium state) $W_A < W_B$

C) If they are compressed by same compression, $W_A = W_B$

D) If they are compressed by same force (upto equilibrium state) $W_A > W_B$

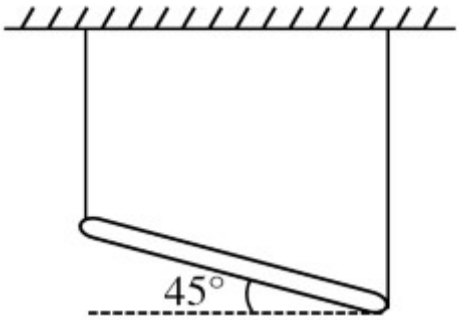
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SECTION - 2 (Maximum Marks : 12)

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Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 26 and 27:

A uniform rod of mass m and length ℓ, is supported at its ends by two vertical strings of unequal length as shown in diagram. Now the right string is burnt. Immediately after burning the string :



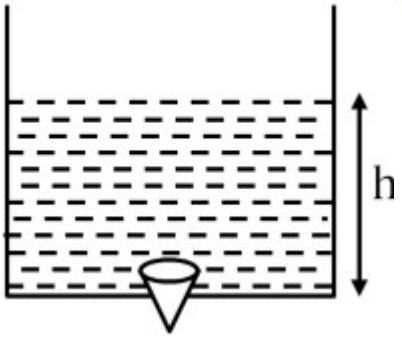
26. The acceleration of centre of the rod is $xg/5$. Find ' x '

27. The tension in the left string is $\frac{x}{5}mg$. Find ' x '

Question Stem for Question Nos. 29 and 29:

A tank has a hole of area $2\ cm^2$ at its bottom and in this hole a conical cork is placed and water is filled up. The water level is made to fall slowly by means of another orifice in the tank. When the water level comes to a height of ' H ' cm , the conical cork just comes out from the hole. Given, volume of water displaced by the cork $=80\ cm^3$, mass of the cork $=40gm$, density of water $=1gm/cm^3$. Area of circular cross-section of base of the conical cork $=8\ cm^2$, Height of cork submerged in water $=\frac{120}{7}$ cm. Take

$g=10\ m/s^2, \pi=\frac{22}{7}, P^0=10^5\ N/m^2$



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28. Calculate the value of H (in cm).

29. Calculate the force exerted by water on the curved surface of the conical cork submerged in water when the liquid level in the vessel is equal to H (in N)

Question Stem for Question Nos. 30 and 31:

A light of wavelength λ is incident on a metal sheet of work function $\phi = 2\text{eV}$. The wavelength λ varies with time as $\lambda = 3000 + 40t$, where λ in Å and t is in second. The power incident on metal sheet is constant at 100 W . This signal is switched on and off for time intervals of 2 minutes and 1 minute respectively. Each time the signal is switched on, the λ start from initial value of 3000Å. The metal plate is grounded and electron clouding is negligible. The efficiency of photoemission is 1% ($hc = 12400\text{eVÅ}$).

30. The time after which photoemission will stop in each cycle is ____ s.

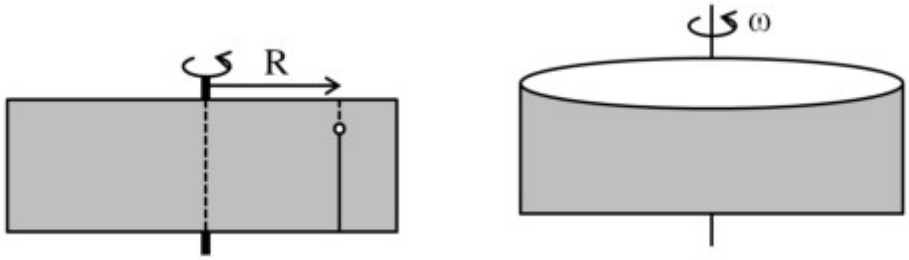
31. The total number of photoelectrons ejected in one hour is ____ $\times 10^{21}$.

SECTION - 3 (Maximum Marks : 12)

This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.
Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
For each question, choose the option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
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Zero Marks: 0 In all other cases.
Negative Marks: –1 In all other cases.

Question Stem for Question Nos. 32 and 33:

A closed cylindrical container with a vertical axis is completely filled with liquid (ρ_w). A plastic bead of density ρ_{body} and volume V is placed at a distance R from the axis and is anchored to the bottom of the container by a thin thread of length ℓ . If as a result of the containers revolutions, the beads sink by h and final distance of bead from axis is r. (In the final state the total content of the container rotates at the same angular speed).



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32. What is force exerted by liquid on the bead?

A) $\rho_w Vg$

B) $\left(\rho_w - \rho_{body}\right)Vg$

C) $\left(\rho_w - \rho_{body}\right)V\sqrt{g^2 + \left(\omega^2 r\right)^2}$

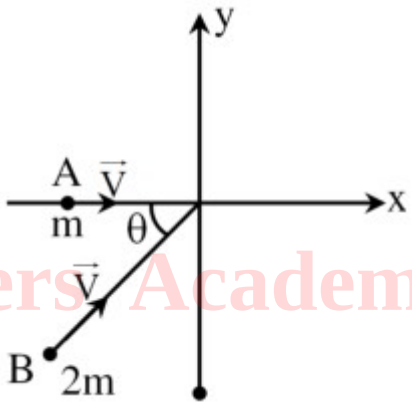
D) $\rho_w V\sqrt{g^2 + \left(\omega^2 r\right)^2}$

33. What is ω

A) $\omega = \sqrt{\frac{(R+r)g}{(\ell+h)r}}$ B) $\omega = \sqrt{\frac{(R-r)g}{(\ell-h)r}}$ C) $\omega = \sqrt{\frac{(R+r)g}{(\ell-h)r}}$ D) $\omega = \sqrt{\frac{(R-r)g}{(\ell+h)r}}$

Question Stem for Question Nos. 34 and 35:

A completely inelastic collision takes place between two body A and B of masses m and 2 m moving respectively with speed v each as shown. The collision is oblique and before collision A is moving along positive x -axis while B is moving at angle θ with x -axis as shown. Then –



34. Speed of composite body after collision is :

A) $\frac{v}{3}\sqrt{5+4\cos\theta}$ B) $\frac{v}{3}\sqrt{5+4\sin\theta}$ C) $\frac{v}{3}\sqrt{4+5\cos\theta}$ D) None of these

35. The angle α that the velocity vector of composite body makes with x -axis after collision is:

A) $\tan^{-1}\left[\frac{2\sin\theta}{1+2\cos\theta}\right]$

B) $\tan^{-1}\left[\frac{2\cos\theta}{1+2\sin\theta}\right]$

C) $\tan^{-1}\left[\frac{2\sin\theta}{1+2\sin\theta}\right]$

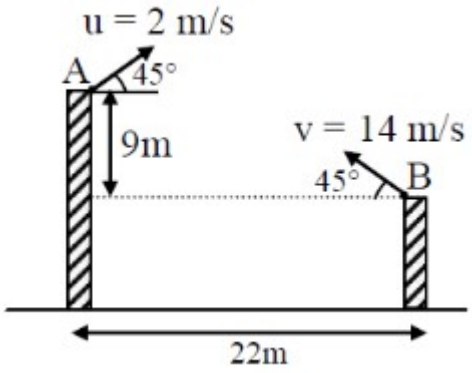
D) None of these

For More Join: @IITJEE_Advanced

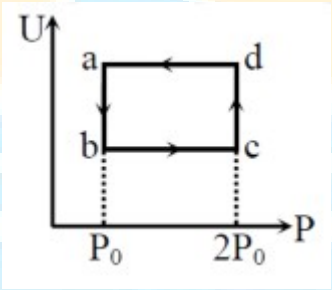
SECTION - 4 (Maximum Marks : 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

36. Two particles are simultaneously thrown from top of two towers as shown. Their velocities are 2 m/s and 14 m/s. Horizontal and vertical separation between these particles are 22 m and 9 m respectively. Then the minimum separation between the particles in process of their motion in meters is ($g=10\ m/s^2$)



37. Figure shows the variation of internal energy (U) with the pressure (P) of 2.0 mole gas in cyclic process abcda. The temperature of gas at c and d are 300 K and 500 K . The net heat exchanged by the gas during the complete cyclic process is given by $K(100)R\ln 2$. Find the value of K .



38. An observer is moving towards a source with speed 15 m/s. The source is moving with speed 5 m/s in the same direction. Air is blowing with speed 10 m/s from observer to source. If frequency of sound emitted is 325 Hz then frequency of sound heard by observer (velocity of sound in air = 330 m/s) in Hz is equal to X . Find the value of $\frac{X-325}{2}$.

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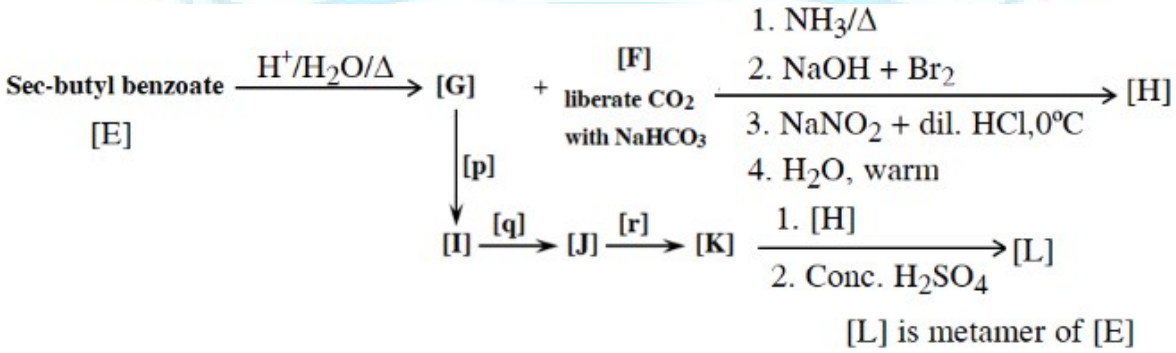
CHEMISTRY **MAX.MARKS: 60**

SECTION - 1 (Maximum Marks : 24)

This section contains **SIX (06)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

39. Choose the CORRECT options regarding the given scheme?



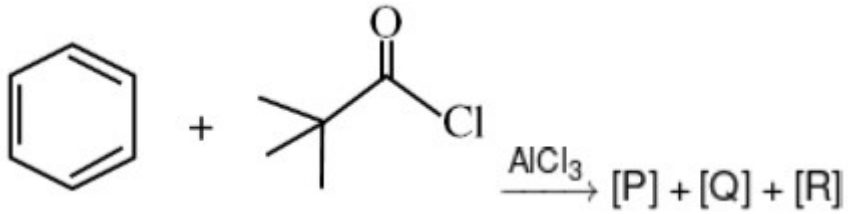
A) $[p]=SOCl_2\,[q]=Mg/\text{dry ether}\,[r]=1.HCHO;2.H^+$

B) $[p]=KBr+H_2SO_4\,[q]=AgCN/DMF\,[r]=H_3O^+/\Delta$

C) $[p]=PBr_3\,[q]=Mg/\text{dry ether}\,[r]=1.CO_2;2.H^+$

D) $[p]=HBr\,[q]=NaCN/DMF\,[r]=H_3O^+/\Delta$

40.

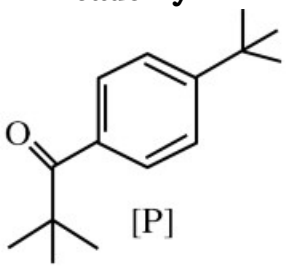


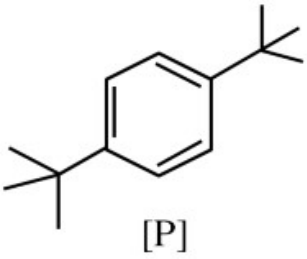
In the above scheme, [P] is the disubstituted aromatic compound and [Q] & [R] are monosubstituted aromatic compounds. [P] and [Q] respond to 2,4 -DNP whereas [R] does not. Choose the correct option(s)

A) [R] is more reactive than benzene towards nitration

B) [R] is more activating than [P] & [Q] towards electrophilic substitution reaction

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C) 

D) 

41. Consider the complex $\left[Fe(en)(gly)(OH)(NH_3)\right]^+$. The CORRECT statement is/are:

A) Coordination number $=6$

B) Oxidation state $=+3$

C) No. of $Fe-O$ bonds $=1$

D) Number of Fe-N bonds $=4$

42. Which of the following is/are CORRECT statement regarding extraction of Iron from haematite ore?

A) CO is the principle reducing agent for hematite ore

B) Fe_2O_3 is used to oxidise impurities in puddling furnace/process

C) Slag formed in metallurgy of iron is $CaSiO_3$

D) Iron obtained from blast furnace is wrought iron

43. Select the CORRECT statement(s) about properties of solid.

A) In antiferromagnetic substance, the magnetic moments of domains are oppositely oriented and cancel out each other's magnetic moment

B) $H_2O, NaCl$ and C_6H_6 are weakly repelled by external magnetic field.

C) CrO_2 is strongly attracted by external magnetic field.

D) Metals conduct electricity in solid as well as in molten state

44. The compound Na_2IrCl_6 reacts with triphenylphosphine in diethyleneglycol in an atmosphere of CO to give $\left[IrCl(CO)(PPh_3)_2\right]$, nown as Vaskas compound. (Atomic number of $Ir=77$) Select CORRECT for Vaskas compound :

A) The IUPAC name of the complex is carbonylchloridobis (triphenylphosphine) iridium(I)

B) The hybridization of the metal ion is sp^3

C) The magnetic moment (spin only) of the complex is zero

D) The complex shows geometrical as well as optical isomerism

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SECTION - 2 (Maximum Marks : 12)

This section contains **TWO (3)** Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +2 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 45 and 46:

The boiling point of a 0.1 molal aqueous barium chloride solution (solution A) is $x^{\circ}C$. To this solution A , an equal volume of 0.2 molal aqueous silver nitrate solution is added to make a new solution B . The difference in the boiling points of the two solutions A and B is $y \times 10^{-2}\,^{\circ}C$.

(Assume : Densities of the solutions A and B are the same as that of water and the soluble salts dissociate completely.)

Use: Molal elevation constant (Ebullioscopic Constant), $K_b = 0.5\ K\ kg\ mol^{-1}$; Boiling point of pure water as $100^{\circ}C$.)

45. The value of x is ____

46. The value of $|y|$ is ____

Question Stem for Question Nos. 47 and 48:

For an electrode reaction written as

$$M^{n+} + ne^- \rightarrow M$$

$$E_{red} = E^{\circ}_{red} - \frac{RT}{nF}\ln\frac{1}{[M^{n+}]} \quad \text{(Nernst equation)}$$

$$= E^{\circ}_{red} - \frac{0.0591}{n}\log\frac{1}{[M^{n+}]} \quad \text{at 298 K}$$

For the cell reaction, $aA + bB \rightleftharpoons xX + yY$

$$E_{cell} = E^{\circ}_{cell} - \frac{RT}{nF}\ln\frac{[X]^x[Y]^y}{[A]^a[B]^b}$$

Standard free energy change $\Delta G^{\circ} = -nE^{\circ}_{cell}F$ where n is the number of electrons transferred in the reaction, per mole of reaction.

Standard free energy change $\Delta G^{\circ} = \frac{-2.303RT}{nF}\log K^{\circ}_{eq}$. Where K°_{eq} is the thermodynamic equilibrium constant at TK. K°_{eq} can be calculated from $E^{\circ}_{cell\ by}$ using the relation,

$$E^{\circ}_{cell} = \frac{0.0591}{n}\log K^{\circ}_{eq}$$

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47. On the basis of information available from the reaction

$$\frac{4}{3}Al + O_2 \rightarrow \frac{2}{3}Al_2O_3; \Delta G^\circ = -827 \text{ kJ mol}^{-1}$$

The minimum emf required to carry out the electrolysis of Al_2O_3 is

$\left(F = 96500 \text{ mol}^{-1}\right)$_____Volts

48. E° for the cell, $Zn\left|Zn^{+2}_{(aq)}\right|\left|Cu^{+2}_{(aq)}\right|Cu$; is 1.1 V at $25^\circ C$. The equilibrium constant for the cell reaction $Zn + Cu^{+2}_{(aq)} \rightleftharpoons Cu + Zn(aq)$ is the order of 10^x. The value of x is____ ____

Question Stem for Question Nos. 49 and 50:

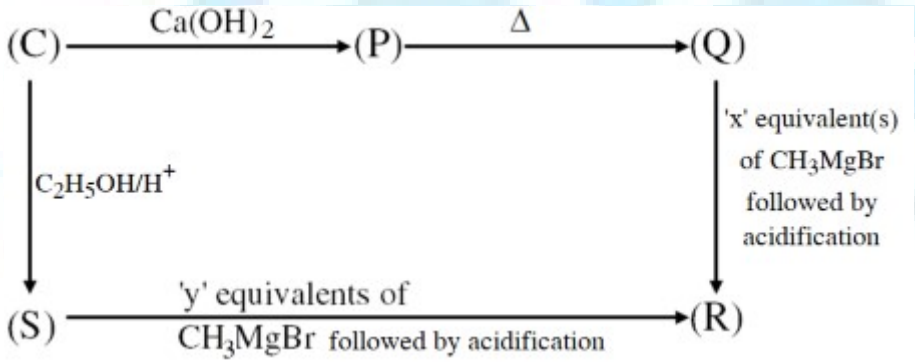
An organic compound (A) $C_3H_6Cl_2$ on reduction with red P and HI gives propane. (A) on hydrolysis by an alkali followed by oxidation gives (B) $C_3H_4O_4$, which on heating gives (C). Both (B) and (C) give effervescence with $NaHCO_3$. (B) on reacting with ethyl alcohol in presence of H^+ gives (D) $C_7H_{12}O_4$, a well known synthetic reagent Now answer the following questions

$$(D) + \text{Benzaldehyde} \xrightarrow[(ii)H_3O^+,\Delta]{(i)\text{Pyridine},\Delta} \text{Major product(E)}$$

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49. The molar mass of major product E is .....(g/mol)
(Report your answer in nearest integer).

50.



' R ' is tert-butyl Alcohol. Find the value of $x + y$.

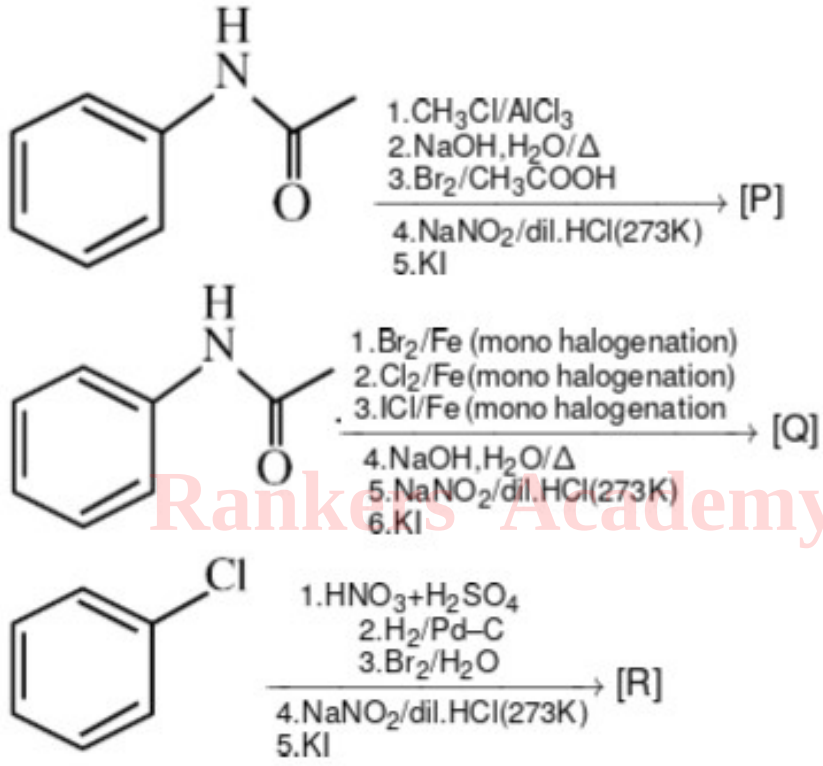
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SECTION - 3 (Maximum Marks : 12)

This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.
Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
For each question, choose the option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +3 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.
Negative Marks: –1 In all other cases.

Question Stem for Question Nos. 51 and 52:

Study the below three schemes and answer the given questions Note consider major organic product in each case and move forward



51. Choose the CORRECT options regarding [P]

A) Sum of the positions of the substituents (as per IUPAC) in [P] is 11 if toluene is taken as a parent chain

B) [P] has plane of symmetry

C) Sum of the positions of the substituents (as per IUPAC) in [P] is 12 if benzene is taken as a parent chain

D) All of these

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52. Choose the INCORRECT options

A) Reaction of [P] with hot $KMnO_4$ followed by decarboxylation with sodalime gives [Q]

B) Reaction of [P] with hot $KMnO_4$ followed by decarboxylation with sodalime gives [R]

C) Reaction of [P] with hot $KMnO_4$ followed by decarboxylation with sodalime gives positional isomer of [Q]

D) All of these

Question Stem for Question Nos. 53 and 54:

In 1886, the French chemist, Francois Raoult after a series of experiments on a number of solvents including water, benzene and ether, succeeded in establishing a relationship between the lowering of vapour pressure or a solution and the mole fraction of the non volatile solute.

$P_1 = X_1 P_1^0$

Where P_1 is vapour pressure of solvent in an ideal solution and P_1^0 is vapour pressure of a pure solvent X mole fraction of solvent and X_2 mole fraction of solute

$X_1 + X_2 = 1$

$P_1 = (1 - X_2) P_1^0$

$\frac{P_1^0 - P_1}{P_1^0} = X_2$

The relative lowering of vapour pressure of a solution containing a non-volatile solute is equal to the mole fraction of solute present in the solution.

53. The vapour pressure of pure liquid X and pure liquid Y are 32 and 102 mm of Hg respectively at 400 K . A liquid mixture of X and Y is composed of 1 mole each. What will be the pressure. when 1 mole of the mixture has been vapourized (round off to nearest integer)

A) 37 B) 57 C) 47 D) 67

54. A solution of non-volatile solute in a mixture of solvent A and Solvent B has a vapour pressure of 40 mm of Hg at $20^\circ C$ and 68 mm of Hg at $30^\circ C$. The vapour pressure of pure solvent (A) is 15 mm of Hg at $20^\circ C$ and 34 mm of Hg at $30^\circ C$ and the vapour pressure of pure solvent (B) is 85 mm of Hg at $20^\circ C$ and 136 mm Hg at $30^\circ C$. Find the mole fraction of solvent A in the solution.

A) 0.20 B) 0.30 C) 0.40 D) 0.50

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SECTION - 4 (Maximum Marks : 12)

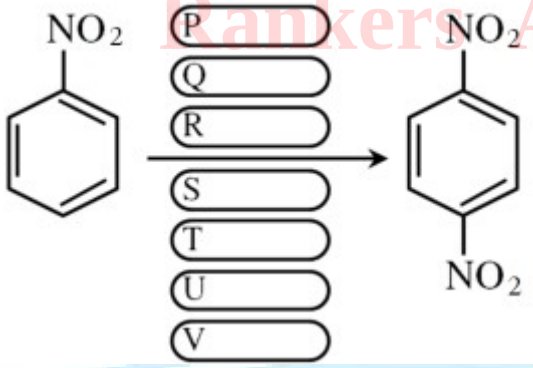
This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

55. Choose the appropriate reagent sequence from the given reagents pool to complete the synthetic conversation given in the questions.

Make sure to use strictly all the boxes given on the arrow, not more not less

In each stage, consider major product.

1. $CH_3Cl/AlCl_3$	5. CuBr/HBr	9. $KMnO_4/OH^-/\Delta$
2. $NaNO_2/Cu/\Delta$	6. CH_3COCl/base	10. $NaNO_2$/dil HCl (273K)
3. dil NaOH/ Δ	7. H_2/Pd-C	11. CuCN/HCN
4. $HNO_3 + H_2SO_4$	8. H_3PO_2/Δ	12. HBF_4
		13. NaOH + CaOH/ Δ



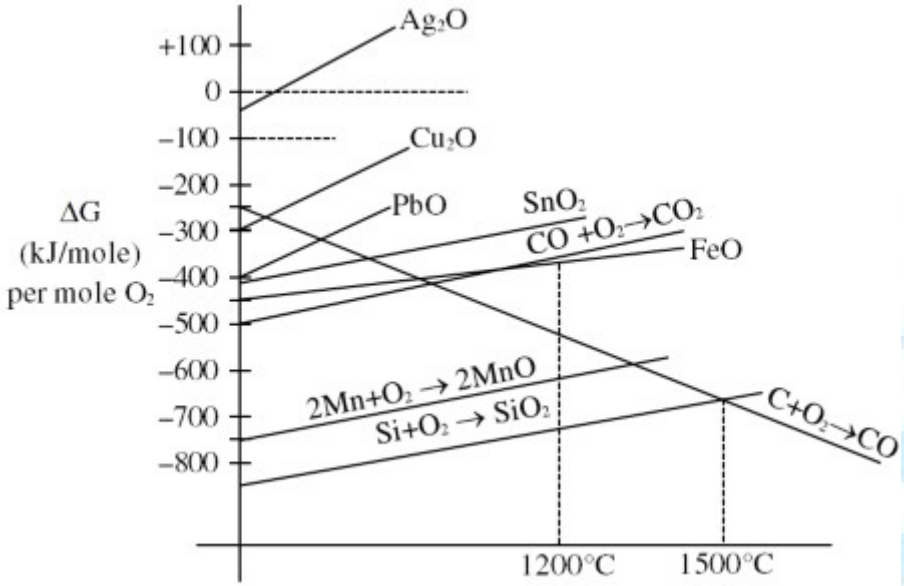
What is the value of $[\mathbf{P}+\mathbf{Q}+\mathbf{R}+\mathbf{S}-\mathbf{T}-\mathbf{U}+\mathbf{V}]$

Note: you can simply mention the numerical values corresponding to the reagents in the boxes P,Q , R, S, T, U & V

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56. How many among the following species CANNOT reduce FeO at temperature range $1200-1500^{\circ}C$ on the basis of given diagram?

Ag, Cu, Pb, Sn, CO, C, Mn, Si



57. The total number of pair of compounds, in which 1st compound is more soluble in water than 2nd compound is

$(BeSO_4 \& BaSO_4), (BeF_2 \& BaF_2), (Na_2CO_3 \& Cs_2CO_3), (CaCrO_4 \& BaCrO_4),$
$(LiCl \& KCl), (NaHCO_3 \& KHCO_3)$

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